AI Tool Prompt Engineering Report

In Partial Fulfillment of the Requirements for:

GE 4120 Elective 3

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Perplexity AI

For this laboratory activity, I chose to work with Perplexity AI, which is essentially a search engine and an AI assistant. It is particularly useful for technical tasks like Electronic Engineering and Circuit Analysis because, unlike some other AI models that might "hallucinate" facts, Perplexity scans the live web to find actual datasheets and component manuals. I selected this tool because accuracy is non-negotiable when you are dealing with wiring components like the ESP32 and SX1278 LoRa module. Having the AI cite its sources, such as a specific manufacturer's pinout, gives me a level of confidence that a standard chatbot just cannot match.

The Prompting Process

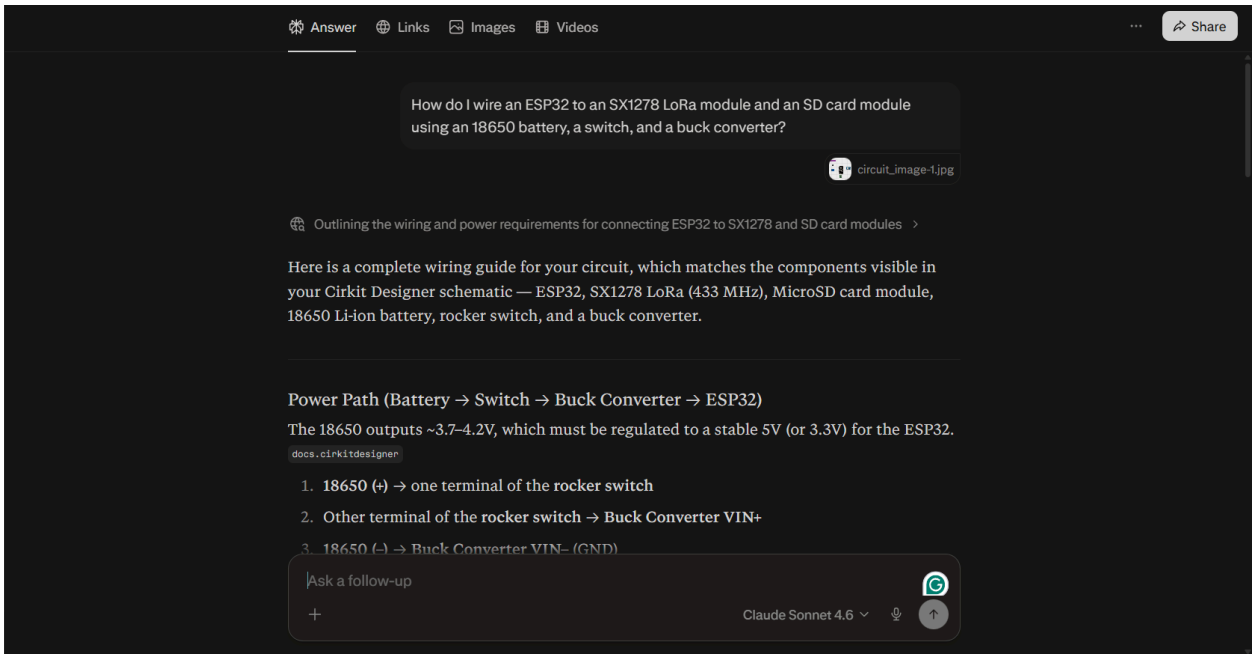

Figure 1.1. Perplexity Initial Prompt

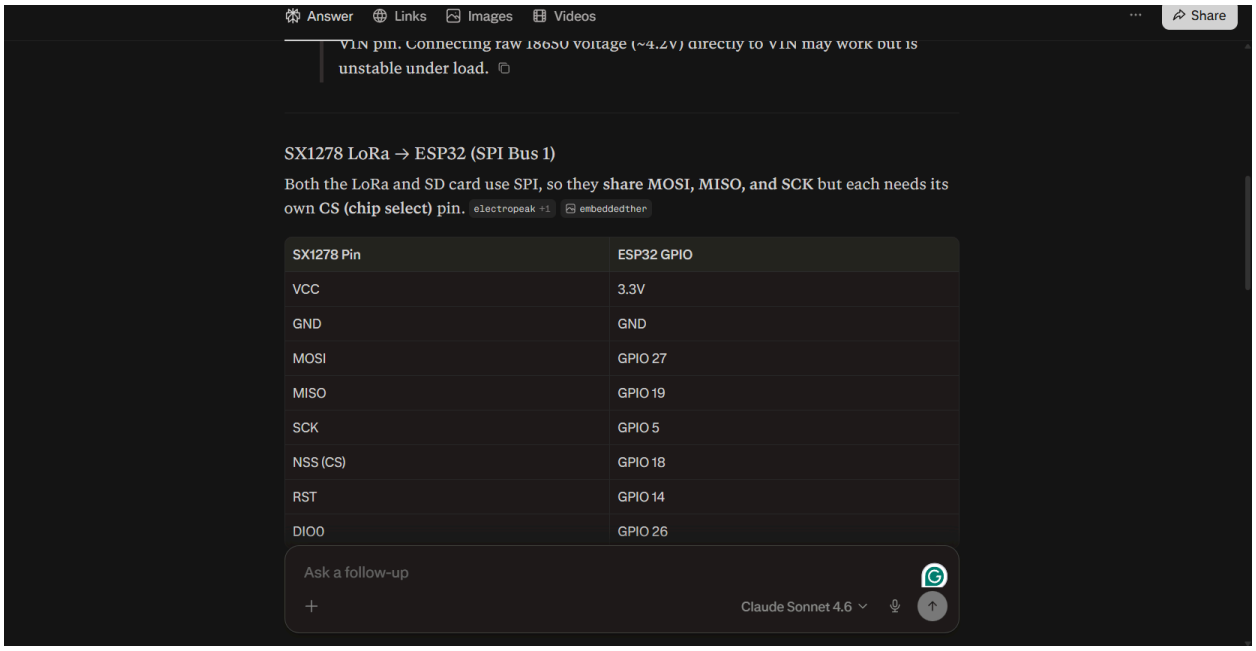

Figure 1.2. Perplexity Initial Prompt

I started with a very straightforward human prompt: "How do I wire an ESP32 to an SX1278 LoRa module and an SD card module using an 18650 battery, a switch, and a buck converter?" This is how most people naturally ask for help, keeping it short and to the point. The result was acceptable, but it felt a bit like a casual blog post. It gave me a simple list of "wire A to pin B" without explaining how to handle the SPI bus sharing between the LoRa and SD modules.

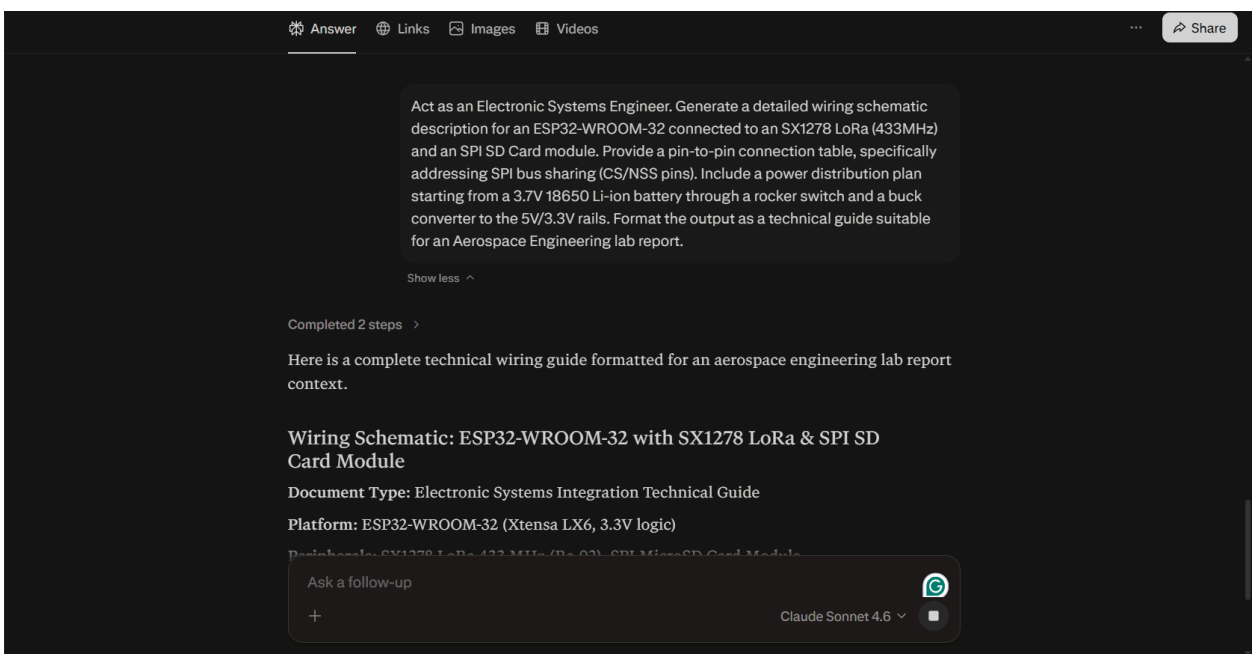

Figure 1.3. Perplexity with Gemini AI-Assisted Prompt

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Wiring Schematic: ESP32-WROOM-32 with SX1278 LoRa & SPI SD Card Module

Document Type: Electronic Systems Integration Technical Guide

Platform: ESP32-WROOM-32 (Xtensa LX6, 3.3V logic)

Peripherals: SX1278 LoRa 433 MHz (Ra-02), SPI MicroSD Card Module

Power Source: 18650 Li-ion (3.7V nominal) → Rocker Switch → DC-DC Buck Converter

1. System Architecture Overview

The system uses a **single SPI bus** shared between the SX1278 LoRa transceiver and the MicroSD card module. SPI bus arbitration is achieved through independent, software-controlled **Chip Select (CS/NSS)** lines — only one peripheral is active at any time. All logic operates at **3.3V**, consistent with ESP32-WROOM-32 GPIO tolerances. The 18650 cell's unregulated output is conditioned through a DC-DC buck converter before reaching the microcontroller. randomnerdtutorials +1

2. Power Distribution Plan

The power chain flows from the raw 18650 cell through protective switching, voltage regulation, and then distribution to each subsystem rail.

2.1 Power Chain Topology

Ask a follow-up

Claude Sonnet 4.6

Figure 1.4. Perplexity with Gemini AI-Assisted Prompt

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Rail	Source	Consumers	Max Current
5V (regulated)	Buck Converter VOUT	ESP32 VIN, SD Module VCC	~1.5A
3.3V (internal)	ESP32 onboard LDO	ESP32 core, SX1278 VCC, SD logic	~600 mA
GND (common)	Buck Converter GND	All modules	—

> ⚠️ **Safety Note:** A single 18650 cell must include over-discharge protection (≥2.5V cutoff). Use a protected cell or integrate a TP4056 with DW01A protection IC to prevent cell damage. arduino +1

3. SPI Bus Configuration

The ESP32-WROOM-32 has two accessible SPI controllers: **HSPI** (SPI2) and **VSPI** (SPI3). This design uses **VSPI** as the primary SPI bus for both peripherals. randomnerdtutorials

3.1 VSPI Bus Pin Assignment (Shared Lines)

SPI Signal	ESP32 GPIO	Direction	Description
SCK	GPIO 18	Output	Serial clock — shared by all SPI devices
MOSI	GPIO 23	Output	Master Out Slave In — shared

Ask a follow-up

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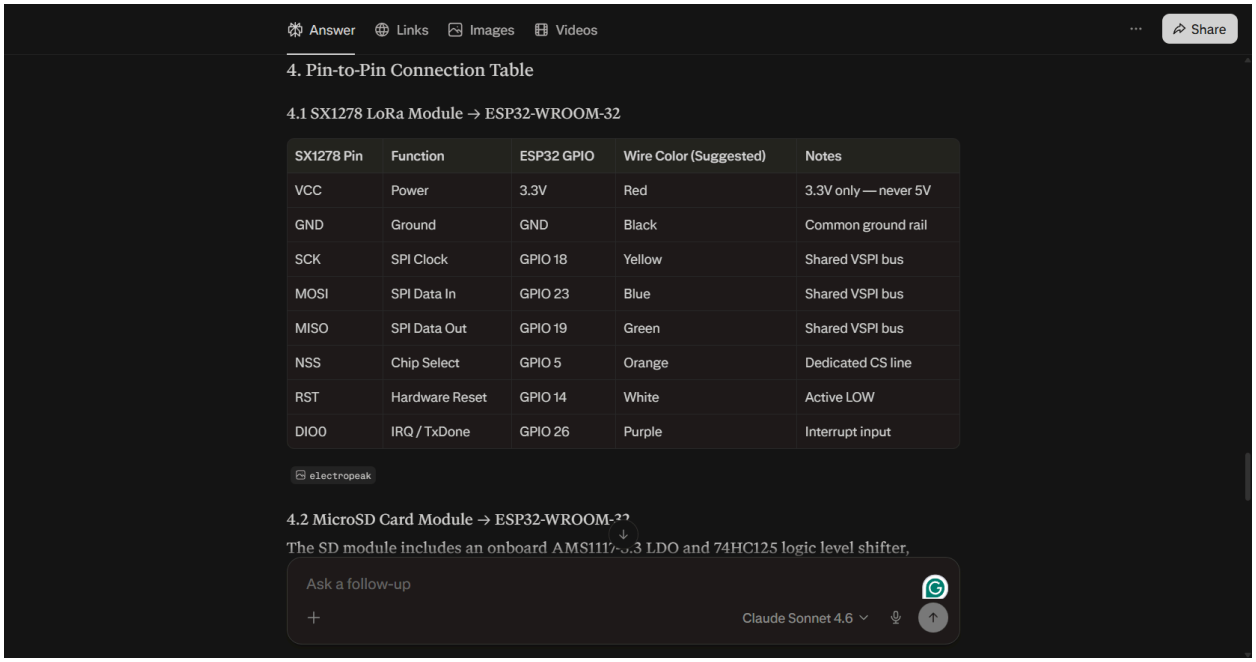
Figure 1.5. Perplexity with Gemini AI-Assisted Prompt



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4. Pin-to-Pin Connection Table

4.1 SX1278 LoRa Module → ESP32-WROOM-32

SX1278 Pin	Function	ESP32 GPIO	Wire Color (Suggested)	Notes
VCC	Power	3.3V	Red	3.3V only — never 5V
GND	Ground	GND	Black	Common ground rail
SCK	SPI Clock	GPIO 18	Yellow	Shared VSPI bus
MOSI	SPI Data In	GPIO 23	Blue	Shared VSPI bus
MISO	SPI Data Out	GPIO 19	Green	Shared VSPI bus
NSS	Chip Select	GPIO 5	Orange	Dedicated CS line
RST	Hardware Reset	GPIO 14	White	Active LOW
DIO0	IRQ / TxDone	GPIO 26	Purple	Interrupt input

electropeak

4.2 MicroSD Card Module → ESP32-WROOM-32

The SD module includes an onboard AMS1117-3.3 LDO and 74HC125 logic level shifter,

Ask a follow-up

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Figure 1.6. Perplexity with Gemini AI-Assisted Prompt

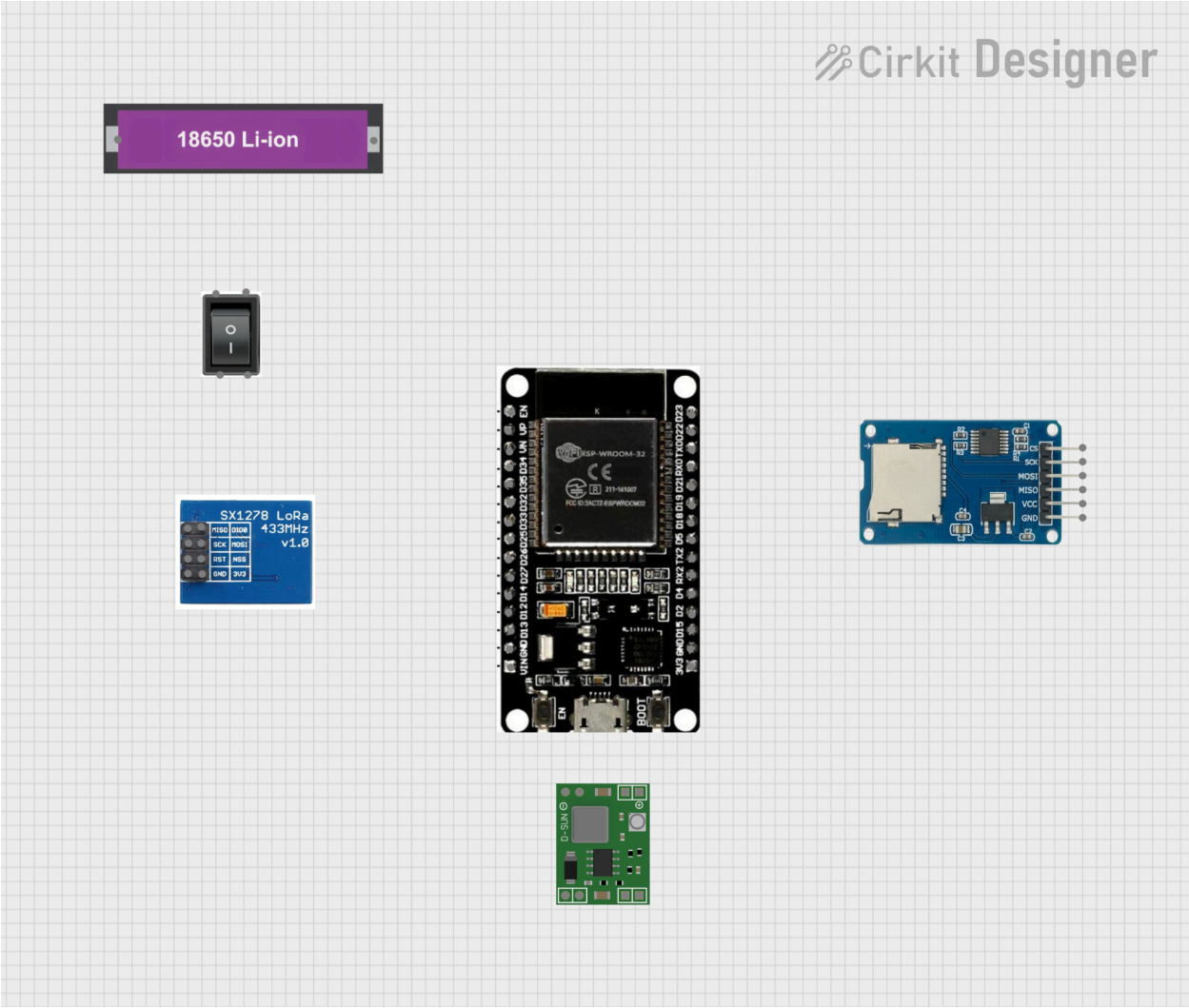
To see how much further I could push the tool, I used Gemini to help me rewrite the request into something a professional might use. The improved version was much more specific: *"Act as an Electronic Systems Engineer. Generate a detailed wiring schematic description for an ESP32-WROOM-32 connected to an SX1278 LoRa (433MHz) and an SPI SD Card module. Provide a pin-to-pin connection table, specifically addressing SPI bus sharing (CS/NSS pins). Include a power distribution plan starting from a 3.7V 18650 Li-ion battery through a rocker switch and a buck converter to the 5V/3.3V rails. Format the output as a technical guide suitable for an Aerospace Engineering lab report."* By giving the AI a persona and a specific format to follow, I was essentially telling it to stop acting like a hobbyist and start acting like an engineer.

Comparison and Analysis

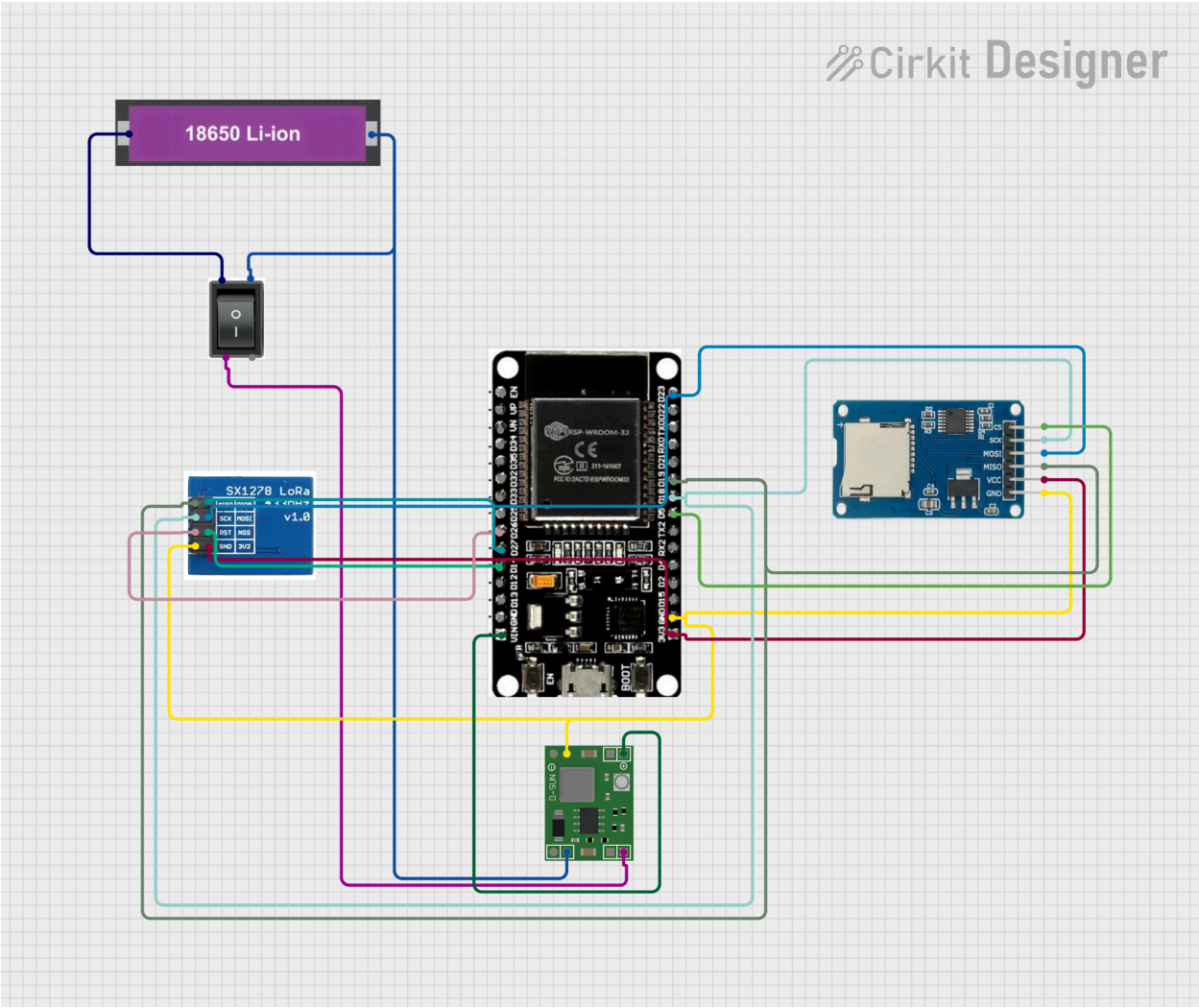
Comparing the two results made it clear that the quality of the AI's "brain" is heavily dependent on how you talk to it. When I gave it the simple human prompt, Perplexity gave me a "good enough" answer. It was fine for a quick DIY project, but it skipped over the tiny details that actually matter in a lab setting, such as the voltage regulation requirements for the ESP32 or the potential interference on the SPI bus when multiple devices are connected. It assumed I wanted the easiest explanation possible, which often means leaving out the technical "why" behind the wiring.

Once I plugged in the Gemini-optimized prompt, the shift in quality was immediate. Because I asked it to act as an engineer and requested a specific power distribution plan, the AI stopped looking at hobbyist forums and started pulling information from more technical sources. It provided a structured, professional guide that detailed exactly how the buck converter should step up or stabilize the battery voltage to ensure the LoRa module did not brown out during transmission.

Appendix



Circuit Diagram without Wiring



Circuit Diagram with Wiring